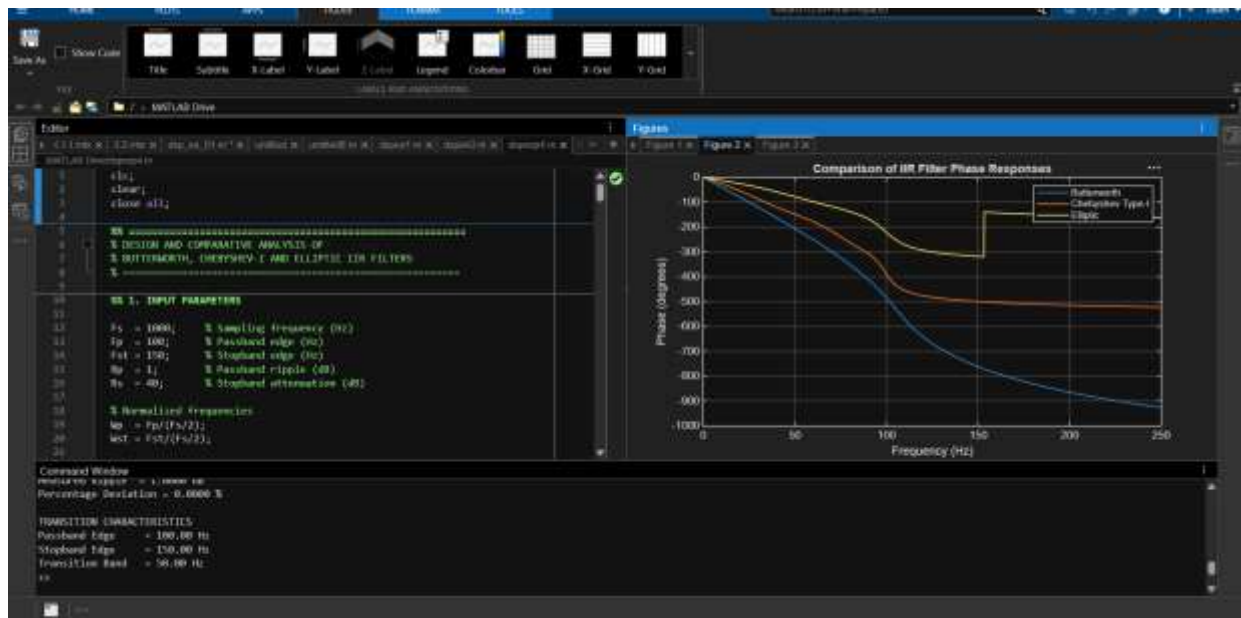
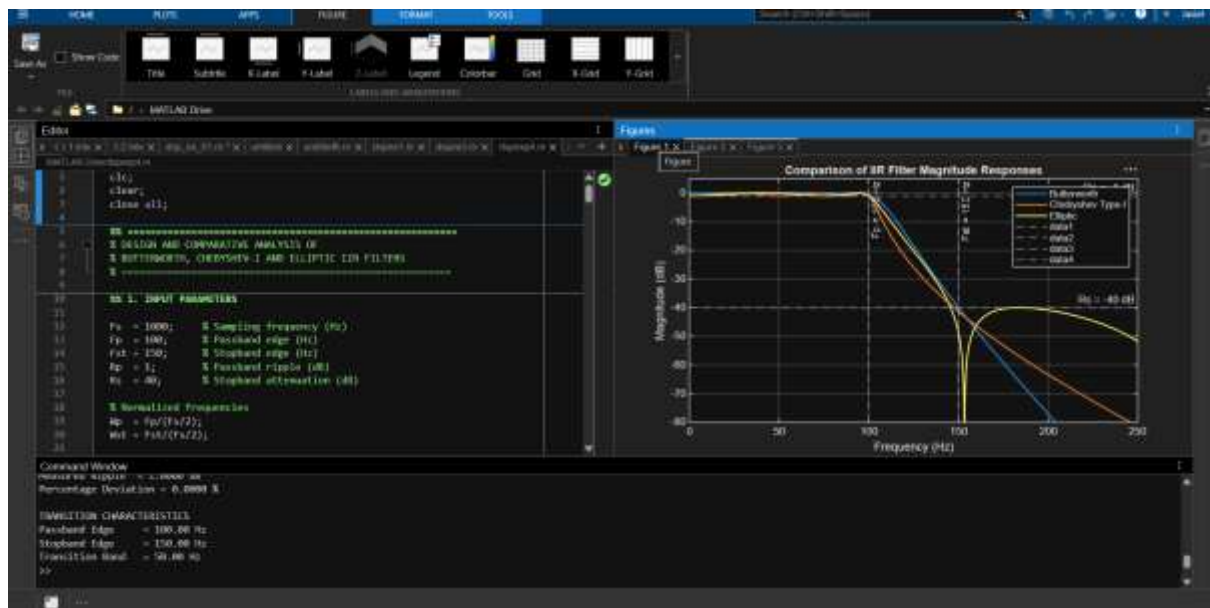
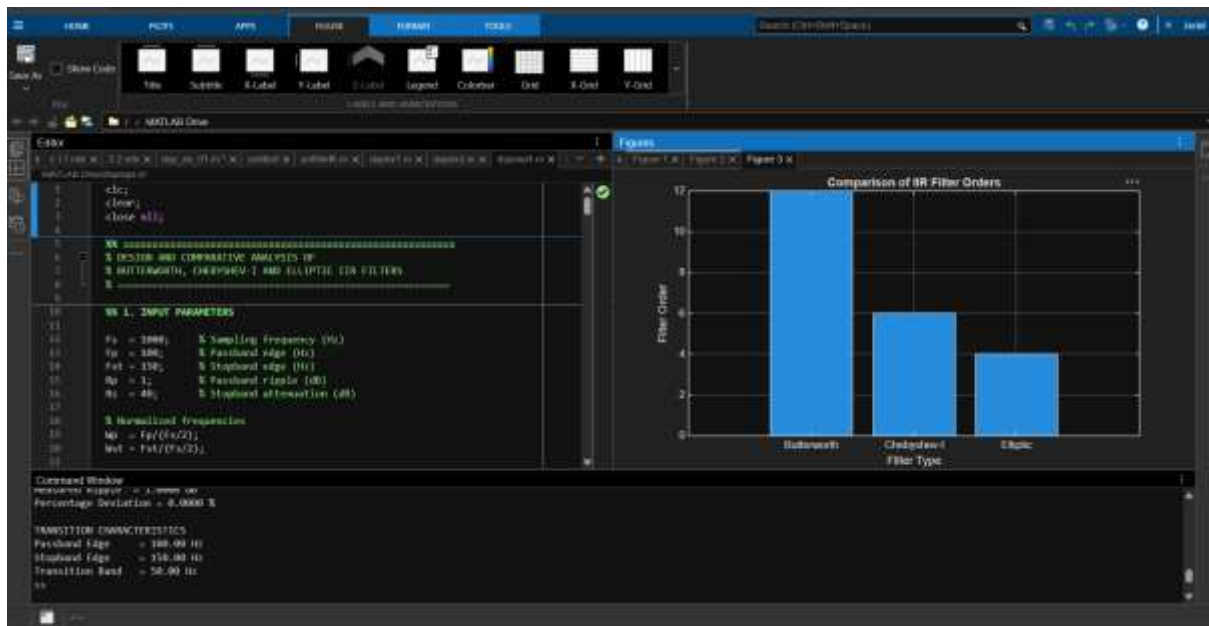






IIR FILTER COMPARATIVE ANALYSIS

INPUT PARAMETERS

Parameter	Value
Sampling Frequency	1000.00 Hz
Passband Edge Frequency	100.00 Hz
Stopband Edge Frequency	150.00 Hz
Passband Ripple	1.00 dB
Stopband Attenuation	40.00 dB

CHEBYSHEV TYPE-I FORMULA

Parameter	Value
Ripple factor epsilon	0.5088
Calculated order	6.2071
Theoretical minimum N	7

FILTER ORDER COMPARISON

Filter Type	Order
Butterworth Order	12
Chebyshev Type-I Order	6
Elliptic Order	4

BUTTERWORTH COEFFICIENTS

Numerator b =

1.0e-03 *

Coefficient	Value
b0	0.0002
b1	0.0027
b2	0.0146
b3	0.0488
b4	0.1098
b5	0.1757
b6	0.2050
b7	0.1757
b8	0.1098
b9	0.0488
b10	0.0146
b11	0.0027
b12	0.0002

```

Denominator a =
  1.0000  -6.8844  22.4341 -45.5100  63.7725 -64.8501  48.9643 -27.6092  11.5218  -3.4662  0.7128  -0.0899  0.0053

=====
CHEBYSHEV TYPE-I COEFFICIENTS
=====
Numerator b =
  1.0e-03 *
  0.0464  0.2782  0.6956  0.9274  0.6956  0.2782  0.0464

Denominator a =
  1.0000  -4.8694  10.3007 -12.3367  8.5969 -3.3268  0.5506

=====
ELLIPTIC COEFFICIENTS
=====
Numerator b =
  0.0197  -0.0171  0.0333  -0.0171  0.0197

Denominator a =
  1.0000  -3.0330  3.8118  -2.2911  0.5554

=====
FILTER PERFORMANCE

```

PASSBAND RIPPLE

Butterworth = 0.8033 dB

Chebyshev Type-I = 1.0000 dB

Elliptic = 1.0000 dB

STOPBAND ATTENUATION

Butterworth = 40.0198 dB

Chebyshev Type-I = 41.3365 dB

Elliptic = 40.0000 dB

CHEBYSHEV RIPPLE DEVIATION

Specified Ripple = 1.00 dB

Measured Ripple = 1.0000 dB

Percentage Deviation = 0.0000 %

TRANSITION CHARACTERISTICS

Passband Edge = 100.00 Hz

Stopband Edge = 150.00 Hz

Transition Band = 50.00 Hz

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